

ZAHEER ABBAS

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING ENGINEER

Data Scientist • Full-Stack & Mobile Developer

Specialization: Applied Machine Learning & AI Systems Engineering

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[GitHub](#) | [LinkedIn](#) | [Hugging Face](#) | [Kaggle](#)

PROFESSIONAL SUMMARY

AI/Machine Learning Engineer specializing in applied ML and end-to-end AI systems — shipping production models, APIs, and deployed applications, not research prototypes. Computer Science graduate of COMSATS University Islamabad (Jan 2026) with 4 years of hands-on project experience and 2 years of freelance/internship delivery across Machine Learning, Deep Learning, NLP, Computer Vision, and full-stack/mobile development. Built and deployed 50+ public repositories of production AI systems, LLM/agentic applications, and cross-platform software, trained and evaluated on datasets from 34,000+ curated text records to 87,000+ real-world rows. Expanding into LLM/agentic engineering (LangChain, LangGraph, CrewAI, RAG, multi-agent orchestration) alongside core ML and full-stack engineering.

KEY ACHIEVEMENTS

- Trained a Hadith-authenticity classifier on a curated corpus of **34,000+ narrations**, deployed behind a three-tier inference fallback (direct model, Hugging Face Inference API, rule-based analysis) with confidence scoring.
- Reached **99.55% weighted F1** across three severity classes with a traffic-congestion classifier evaluated on **87,000+ real-world rows** collected across 12 Pakistani cities.
- Achieved **99.97% test accuracy** and a **0.9997 weighted F1-score** with a diet/health recommendation model, deployed as a public Hugging Face Space.
- Shipped and maintain **45+ public repositories** spanning ML, LLM/agentic AI, and full-stack/mobile engineering — each delivered as working, documented, deployed code.
- Sustained continuous freelance delivery since 2024 across **three platforms** (Upwork, Freelancer.com, PeoplePerHour), independently owning each client engagement end-to-end.

TECHNICAL SKILLS

Machine Learning / Deep Learning: scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Gradient Boosting, Random Forest, LSTM, GRU, ARIMA/GARCH, Ensemble Learning, Transfer Learning, Reverse-Mode Automatic Differentiation, Model Evaluation & Cross-Validation

NLP & LLM Engineering: Hugging Face Transformers, AraBERT, BERT, Sentence-Transformers, spaCy, TextBlob, Prompt Engineering, Retrieval-Augmented Generation (RAG), Hybrid Retrieval (BM25 + Dense Embeddings + Reciprocal Rank Fusion), LangChain, LangGraph, CrewAI, Multi-Agent Systems, Typed Tool Calling, Synthetic Data & DPO Curation

Languages: Python, SQL, TypeScript, JavaScript, Dart, Kotlin

Computer Vision: YOLOv8, OpenCV, MobileNetV2, VGG16, Grad-CAM, Image Classification, Real-Time Object Detection, Explainable AI

Backend & APIs: Flask, FastAPI, Uvicorn, SQLAlchemy, Pydantic, JWT Authentication, REST API Design, Swagger/OpenAPI Documentation

Data & DevOps: Pandas, NumPy, EDA, Feature Engineering, MongoDB, SQLite, PostgreSQL-ready, Docker, Nginx, Git/GitHub, Hugging Face Hub

Frontend & Mobile: React 19, Vite, Tailwind CSS, Recharts, Flutter, Dart, Kotlin, Jetpack Compose, Riverpod, Provider, Go Router, Firebase (Auth/Firestore/Cloud Messaging), Cloudinary

Security & Systems: Bluetooth Mesh Networking, ECDH Key Exchange, AES-256-GCM Encryption, PII Sanitization, Privacy-Aware Pipelines (GDPR/CCPA-aligned)

Maps & Real-Time: Google Maps Platform, Geolocation & Geocoding, Socket.IO, Gradio API Integration

Additional Certified Skills: Graphic Design (Canva, Adobe Photoshop, Figma), E-Commerce Management, Search Engine Optimization (on-page/off-page)

EXPERIENCE

PROFESSIONAL EXPERIENCE

AI/ML Intern — Certura Technologies

Jun 2025 – Jul 2025

Remote

- Built and shipped three applied NLP/CV tools during a two-month internship: an NLP-based FAQ chatbot, an image classifier (MobileNetV2) with bulk ZIP-upload support, and a text summarization tool (spaCy + TextBlob), each delivered as a working Streamlit application.
- Worked within a live team environment on real deliverables and deadlines, gaining direct exposure to end-to-end applied ML delivery rather than isolated coursework projects.

Freelance AI/ML & Data Services — Upwork · Freelancer.com · PeoplePerHour

2024 – Present

Remote

- Delivered end-to-end machine learning, data processing, and technical writing engagements for international clients across three freelance platforms — Upwork, Freelancer.com, and PeoplePerHour — sustaining an active, continuous client track record since 2024.
- Independently scoped, priced, and executed each engagement solo, covering the full delivery lifecycle: requirements gathering, data cleaning, model training, API development, and Hugging Face deployment, then carried those production practices directly into personal portfolio projects.

Ambassador — Paklaunch

May 2024 – Present

Pakistan · Certificate of Internship (May 5 – Aug 2, 2024)

- Completed a certified three-month internship with Paklaunch, recognized for outstanding contribution, dedication, and passion, before continuing in an ongoing Ambassador capacity.
- Represent Paklaunch's tech community programs, supporting outreach and mentorship activities alongside academic and project work.

PROJECT EXPERIENCE

AI Hadith Authentication System

University Final Year Project

Independent Project | GitHub · Hugging Face Space | MIT License

Tech Stack: Flask, MongoDB, SQLite, Hugging Face Inference API, OCR, ASR, Bcrypt, Docker, Nginx

- Designed and built a full-stack Hadith authenticity classification platform (Flask, MongoDB with SQLite fallback) predicting whether a Hadith is Sahih, Hasan, Da'if, or Mawdu' using a Hugging Face-hosted model trained on a curated corpus of 34,000+ narrations.
- Engineered a three-tier inference fallback (direct model, then Hugging Face Inference API, then rule-based keyword analysis) with confidence scoring and Isnad (chain-of-narration) source attribution.
- Added multi-modal input support — OCR for image-based text and ASR for audio queries — a bilingual (Arabic/English) interface, a Quran-reader module, and Sunnah.com API integration, secured with Bcrypt password hashing and session-based authentication.
- Containerized the full application (Dockerfile, docker-compose, Nginx) for reproducible deployment.

AI Urban Nexus — Smart City Mobile Platform

University Final Year Project

Independent Project | GitHub · Hugging Face Spaces | MIT License

Tech Stack: Flutter, Dart, XGBoost, LSTM, Hugging Face, Firebase (Realtime DB, FCM), Google Maps Platform

- Built a cross-platform Flutter mobile application integrating three independent Hugging Face-hosted machine learning backends: a crime-detection model for stolen-vehicle image recognition, an XGBoost traffic-congestion classifier, and an LSTM-based energy-demand forecaster.
- Traffic-congestion classifier reached 99.55% weighted F1 across three severity classes on an 87,000+ row evaluation set built from real-time data collected across 12 Pakistani cities.
- Implemented Google Maps-based safe-route planning with crime- and traffic-aware scoring, Firebase Realtime Database city statistics, Firebase Cloud Messaging alerts, and a role-gated admin analytics dashboard (traffic officer, security lead, energy manager).
- Designed an offline heuristic fallback layer so core traffic and safety features remain usable when the Hugging Face inference API is unreachable.

Personal Financial Tracker — Full-Stack Web App

Independent Project | GitHub

Tech Stack: FastAPI, SQLAlchemy, Pydantic, JWT, React 19, TypeScript, Vite, Tailwind CSS, Recharts

- Built a full-stack personal finance management app: FastAPI + SQLAlchemy + Pydantic backend with JWT authentication (bcrypt, 12-round hashing) and a React 19 + TypeScript + Vite frontend styled with Tailwind CSS.
- Implemented income/expense tracking, budget alerts (80% warning / 100% exceeded thresholds), savings goals, and 12-month analytics dashboards using Recharts, with auto-generated Swagger API documentation.

Health Hub — AI Nutrition & Diet Management Platform

Independent Project | GitHub

Tech Stack: React 18, Vite, Tailwind CSS, Groq API (Llama 3.3 / Qwen), Firebase (Auth, Firestore)

- Developed a full nutrition-management web platform for Pakistani users with React 18, Vite, and Tailwind CSS, covering AI-generated diet plans, lab-report analysis, recipe discovery, and nutritionist appointment booking.
- Integrated the Groq API (Llama 3.3 / Qwen models) through a custom rate-limited client enforcing a five-request-per-minute queue with exponential backoff, generating multi-day diet plans one day at a time to avoid token-limit failures.
- Built role-based access across three roles (patient, nutritionist, admin) on Firebase Authentication and Firestore, with composite query indexes for appointment scheduling and professional PDF report generation for diet plans and lab analyses.

Salon Booking App — Cross-Platform Flutter App

Independent Project | GitHub | MIT License

Tech Stack: Flutter, Dart, Firebase (Auth, Firestore, Storage, FCM), Riverpod, Provider, fl_chart

- Built a Flutter salon-booking application with Firebase Authentication (email/password, OTP, password reset), calendar-based appointment booking, staff assignment, and payment handling across 25+ routes.
- Implemented owner and admin dashboards with revenue charts (fl_chart), Riverpod and Provider state management, and Firebase Firestore, Storage, and Cloud Messaging for real-time data and notifications.

Events Management — Cross-Platform Flutter App

Independent Project | GitHub | MIT License

Tech Stack: Flutter, Dart, Firebase (Auth, Firestore), Cloudinary, Material Design

- Developed a Flutter/Dart event-management application supporting event creation, registration, and real-time updates via Firebase Authentication and Cloud Firestore.
- Implemented Cloudinary-based image uploads and a Material Design UI across six platform targets (Android, iOS, Web, Windows, macOS, Linux) from a single codebase.

Chat-boot — AI-Powered FAQ Chatbot

Independent Project | GitHub

Tech Stack: Python, Sentence-Transformers, Flask, Streamlit, Ngrok

- Built a semantic-search FAQ chatbot using Sentence-Transformers for embedding-based question matching, exposed via a Flask REST API and an interactive Streamlit dashboard.
- Structured a modular three-part chatbot engine (FAQ matching, model loading, utilities) with a custom Q&A dataset, and enabled public access testing via Ngrok tunneling.

Health & Diet Recommendation Classifier

Independent Project | Hugging Face Space

Tech Stack: Python, Scikit-learn (Gradient Boosting, Random Forest, Logistic Regression), Hugging Face

- Trained and benchmarked three classifier families — Gradient Boosting, Random Forest, and Logistic Regression — for personalized diet and health recommendations as a standalone machine learning prototype.
- Best-performing model (Gradient Boosting) achieved 99.97% test accuracy and a 0.9997 weighted F1-score on held-out test data; deployed as a public interactive Hugging Face Space.

ADDITIONAL INDEPENDENT PROJECTS (45+ REPOSITORY PORTFOLIO)

35+ additional public repositories spanning applied ML/DL research, generative AI/agentic systems, and full-stack/mobile engineering:

Applied Machine Learning & Deep Learning

- **AI Kidney Tumor Deep Classifier** — Transfer-learning (VGG16) classifier for kidney-tumor detection from CT images with Grad-CAM interpretability and a Streamlit inference interface [*Python, TensorFlow/Keras, VGG16, OpenCV, Grad-CAM*]

- **AI Gold Price Forecasting System** — Ensemble forecasting platform for gold prices using macroeconomic features, with a Flask REST API and scenario/stress-testing workflows *[Python, Pandas, Scikit-learn, XGBoost, Flask]*
- **Market Forecast Ensemble** — Multi-model time-series and quantitative trading engine combining GRU, gradient boosting, and ARIMA/GARCH with walk-forward validation and back-testing *[Python, GRU, Gradient Boosting, ARIMA, GARCH]*
- **ML From Scratch Plus** — Machine learning framework built from first principles in pure NumPy, including a reverse-mode automatic differentiation engine and backpropagation *[Python, NumPy]*
- **Transformer Internals Lab** — Mechanistic interpretability toolkit for Transformer language models: attention-head visualization, residual-stream tracing, and Logit Lens analysis *[Python, PyTorch, Transformers]*
- **AI SmartATS Resume Optimizer** — ATS resume-optimization platform using TF-IDF and cosine similarity for keyword/skill-gap matching against job descriptions, with AI-assisted cover-letter generation *[Python, Gemini, NLP, TF-IDF]*

Generative AI, LLMs & Agentic Systems

- **Hybrid-RAG-Eval** — Hybrid retrieval engine (BM25 + dense embeddings + Reciprocal Rank Fusion) with an evaluation framework scoring context precision/recall and faithfulness for RAG pipelines *[Python, BM25, Dense Embeddings, RAG Evaluation]*
- **AgentMesh** — Multi-agent coordination framework with typed, schema-validated tool dispatch for reliable LLM-to-tool execution *[Python, Multi-Agent Systems, JSON Schema, OpenAPI]*
- **VaultGuard** — Fail-closed PII sanitization agent for AI pipelines, designed around GDPR/CCPA privacy requirements *[Python, AI Security, PII Detection]*
- **DataForge-LLM** — Synthetic instruction-dataset generation and curation pipeline producing DPO-ready preference pairs for LLM post-training *[Python, LLMs, Synthetic Data, DPO]*
- **Sharia Finance Assistant** — Production-oriented Islamic-finance RAG assistant with enterprise PII protection and structured observability *[Python, RAG, LLMs, Observability]*
- **Aegis Code Linter** — AI code-review agent evaluating code against OWASP Top 10 and PEP 8 using LangChain-driven analysis *[Python, LangChain, LLMs, OWASP]*
- **QueryForge** — Natural-language-to-SQL business-intelligence agent for logistics and supply-chain use cases *[Python, LLMs, SQL, Agentic AI]*
- **ScholarSync** — Academic research RAG assistant extracting methodology, findings, and research gaps from PDF papers *[Python, RAG, LLMs, PDF Processing]*
- **MinuteMind / TriageAI / TalentScan** — Enterprise meeting-intelligence, SRE issue-triage, and recruitment-screening agents producing structured, prioritized outputs *[Python, Agentic AI, NLP, GitHub APIs]*
- **SocraticAI / MarketMind / Sentinel Market Intelligence** — Multi-agent deliberation, competitive-intelligence, and go-to-market analysis systems using LangGraph and CrewAI orchestration *[Python, LangGraph, CrewAI, LLMs]*
- **Launchpad / ChefIQ / WayfareAI / AlphaResearch / DataLens / BriefingBot / CorrespoAI / NexusDesk / VoiceAmplify** — Additional agentic and generative-AI systems covering career strategy, culinary intelligence, travel planning, equity research, e-commerce analytics, geopolitical briefings, executive communication, customer-support intelligence, and B2B content generation *[Python, LangChain, CrewAI, Generative AI]*

Full-Stack, Mobile & Systems Engineering

- **Meshline Android** — Offline-first, encrypted Bluetooth Mesh messenger and Push-to-Talk app for connectivity-denied environments *[Kotlin, Jetpack Compose, ECDH, AES-256-GCM]*
- **DocSynth / TestForge** — AI-assisted developer tooling generating Google-style Python docstrings and automated pytest suites for coverage *[Python, Generative AI, Pytest]*
- **Autonomous Agent Framework** — Framework for building autonomous AI agents covering task decomposition and agent-driven execution workflows *[Python, LLMs, Agentic AI]*
- **Portfolio** — Personal developer portfolio presenting AI/ML, Flutter, and web engineering work *[TypeScript, Web Development]*

Full source code and live demos available on GitHub and Hugging Face (linked above).

EDUCATION

Bachelor of Science, Computer Science <i>COMSATS University Islamabad</i> • Relevant coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Systems, Software Engineering	Jan 2026
Intermediate, Pre-Engineering <i>Government Degree College, Bagan, Lower Kurram, Khyber Pakhtunkhwa</i>	2016 – 2018
Matriculation, Science <i>Al Madina Public High School, Bagan, Lower Kurram, Khyber Pakhtunkhwa</i>	2014 – 2016

CERTIFICATIONS

- Intro to Machine Learning — Kaggle, Aug 2026
- Time Series — Kaggle, Aug 2026
- E-Commerce Management, Search Engine Optimization & Graphic Design — DigiSkills.pk (Ignite National Technology Fund / VU Pakistan), 2022–2024 — IDs 236UUMEMK, ZPSC8UMK5, 37GA3JMK5
- Electrician, General Journeyman Apprentice (Inside)
- AI & Machine Learning Internship — Certura Technologies, Jun–Jul 2025
- Certificate of Internship — Paklaunch, May 5 – Aug 2, 2024

LANGUAGES

English (Full Professional) • Urdu (Native) • Pashto (Native) • Punjabi (Conversational)